

Variable	Units	Equation	Documentation
Complex Performance Indicator CPI			
CPI	unitless	gths[2]*Socl+CPIWeights[3]*E	Complex performance indicator
CPIWeights[1]: Environmental	unitless	0.3375	
CPIWeights[2]: Social	unitless	0.5035	
CPIWeights[3]3: Economic	unitless	0.1591	
Economic KPIs			
A	US Dollars	IA+TIME*AER	Assets
EcoBenchs[1]	unitless	0	
EcoBenchs[2]	unitless	0	
EcoWeights[2]	unitless	0.486486486	
EcoWeights[1]	unitless	0.513513514	
ROAMT	months	12	ROA's Measurement Time
MaxEcoKPI[Económico]	unitless	100	
AER	US Dollars/months	1352079	Asset's Exchange Rate
IA	US Dollars	276492461	Initial Assets
EcoKPIsW[Económico]	unitless	IF (EcoBenchs < 0) THEN (	
EcoKPIs[1]	unitless	IF(EcoWeights >0) THEN (	EcoKPI1: Gross Profit
EcoKPIs[2]	unitless	IF(EcoKPIs<=EcoBenchs)	EcoKPI2: Return on assets
EBIT	US Dollars/months	THEN(	
Ecol	unitless	EcoWeights/SUM(ABS(EcoW	
		eights))*(EcoKPIs/EcoBenchs	
		)*100) ELSE(	
		SR , 2) *100	
		2)* 100	
		SR-TPC-WCAC-FCEC-ODC	Earnings before interest and taxes
		F(EcoWeights<0) THEN -	
Environmental KPIs			
AEFS	Tons/months	7.858187741	Average Emission from sector
ECU	US Dollars/unidades	17.21	Energy Consumption per Unit
EnviBenchs[1]	unitless	0	
EnviBenchs[2]	unitless	0	
EnviBenchs[3]	unitless	0	
EnviBenchs[4]	unitless	0	
EnviBenchs[5]	unitless	0	
EnviWeights[1]	unitless	0.185676393	
EnviWeights[2]	unitless	-0.209549072	
EnviWeights[3]	unitless	-0.188328912	
EnviWeights[4]	unitless	-0.204244032	
EnviWeights[5]	unitless	-0.212201592	
MaxEnvKPI[Ambiental]	unitless	100	
WTC	US Dollars/Unidad	2600	Waste Treatment Cost
EnviKPIs[1]	unitless	(IRM+DRM)/(IMC+DMC)*100	recycled materials and raw materials.
Envil	unitless	F(EnviWeights<0) THEN -	
EnviKPIs[2]	unitless	TEC//TPC*100	EnviKPI2: Fuel consumption.
EnviKPIs[3]	unitless	TW // APr *100	EnviKPI3: Waste production
EnviKPIs[4]	unitless	TCWD // SR *100	EnviKPI4: Environmental costs
EnviKPIs[5]	unitless	Emi // AEFS *100	greenhouse gases.
EnviKPIsW[Ambiental]	unitless	IF (EnviBenchs < 0) THEN (	
TCWD	US Dollars/months	IF(EnviWeights >0) THEN (	
TEC	US Dollars/months	IF(EnviKPIs<=EnviBenchs)	
		THEN(	
		EnviWeights/SUM(ABS(Envi	
		Weights))*(EnviKPIs/EnviBen	
		chs)*100) ELSE(	
		WP*WTC	Total Cost of Waste Disposal
		Pr*ECU	Total Energy Consumption
Social KPIs			
MaxSocKPI[1]	unitless	100	

MaxSocKPI[2]		100	
MaxSocKPI[3]		200	
MaxSocKPI[4]		100	
SocBenchs[1]	unitless	0	
SocBenchs[2]		0	
SocBenchs[3]		100	
SocBenchs[4]		0	
SocWeights[1]	unitless	0.210332103	
SocWeights[2]		-0.258302583	
SocWeights[3]		-0.239852399	
SocWeights[4]		-0.291512915	
SocI	unitless	F(SocWeights<0) THEN -	
SocKPIs[1]	unitless	WCAC // TSC *100	employees covered by collective agreement.
SocKPIs[2]		ODC // TSC * 100	SocKPI2: Occupational diseases.
SocKPIs[3]		((MSC-WSC)//TSC+1)*100	SocKPI3: Wage discrimination.
SocKPIs[4]		FCEC // TSC *100	of Ethics.
		IF (SocBenchs < 0) THEN	
		IF(SocWeights >0) THEN (	
		IF(SocKPIs<=SocBenchs)	
		THEN(	
		SocWeights/SUM(ABS(SocW	
		eights))*(SocKPIs/SocBenchs	
SocKPIsW[Social]	unitless	)*100) ELSE(	
Production sub-model			
AOT	months	12	Average Order Time
NOr	Per Month	0.5	New Orders
TCS	months	0.5	Time to Correct Stock
DeITi	months	1	Delivery Time
DI	Per Month	1	Demand Increase
DMCT	Months	1	Direct Materials Coverage Time
DSCT	months	1	Desired Stock Coverage Time
IMCT	months	1	Indirect Material Coverage Time
MRDM	unitless	1	Materials
MRIM	unitless	1	Materials
PrT	months	1	Production time
STi	months	1	Sales time
TF	Months	1	
TCDM	months	2	Time to Correct Direct Materials
TCIM	months	2	Time to Correct Indirect Materials
PDRM	unitless	5	Materials
WUP	unitless	5	Waste per Unit Produced
PIRM	unitless	10	Materials
PWU	Per Month	17	Percentage of Waste Used
PCU	US Dollars/Unidad	520	Production Cost per Unit
SVU	US Dollars/Unidad	780	Sales Value per Unit
DMCr	Unidades/months	(DDM-DPM)/TCDM	Direct Materials Corrector
IMCr	Unidades/months	(DIM-IPM)/TCIM	Indirect Materials Corrector
SC	Unidades/months	(DS-St)/TCS	Stock Corrector
RPTC	unitless	(SR-TPC)/TPC	Return on production total cost
Cum_RPTC(t)	unitless	(RPTC_Rate) * dt	State of GP
DS	Unidades	D*DSCT	Desired Stock
SR	US Dollars/months	De*SVU	Sales Revenue
DM(t)	Unidades	DMD) * dt	Direct Materials on the Way
DMP	Unidades/months	DMO*(1-PDRM/100)	Direct Materials Purchased
DRM	Unidades/months	DMO*PDRM/100	Direct Recycled Materials
DDM	Unidades	DP*DMCT*MRDM	Desired Direct Materials
DIM	Unidades	DP*IMCT*MRDM	Desired Indirect Materials
DPM(t)	Unidades	dt	Direct Production Materials

D	Unidades/months	ED*DI	Demand
RPTC_Rate	Per Month	RPTC/DT ELSE 0	Change's Rate of GP
PrR	Unidades/monthsnths	ELSE 0	Production Rate
IM(t)	Unidades	* dt	Indirect Materials on the Way
NIM	unidades/months	IMO*(1-PIRM/100)	New Indirect Materials
IRM	unidades/months	IMO*PIRM/100	Indirect Recycled Materials
IPM(t)	Unidades	IPM(t - dt) + (IMD - IMC) * dt	Indirect Production Materials
DP	Unidades/months	MAX(D+SC, 0)	Desired Production
DMD	Unidades/months	MAX(DM/DelTi, 0)	Direct Materials Delivery
DMO	Unidades/months	MAX(DP*MRDM+DMCr, 0)	Direct Materials Orders
IMO	Unidades/months	MAX(DP*MRIM+IMCr, 0)	Indirect Material Orders
IMD	Unidades/months	MAX(IM/DelTi, 0)	Indirect Materials Delivery
Pr	Unidades/months	IPM),MIN(DPM/MRDM,	Production
De	unidades/months	MIN(St/STi, D)	Deliveries
Ords(t)	Unidades	Ords(t - dt) + (NOrI) * dt	Orders
NOrI	Unidades/months	Ords*NOr/100	New Orders Increase
TPC	US Dollars/months	PC+TSC+TCWD	Total Production Cost
DMC	Unidades/months	Pr*MRDM	Direct Materials Consumption
IMC	Unidades/months	Pr*MRIM	Indirect Materials Consumption
PC	US Dollars/months	Pr*PCU	Production Costs
WP	Unidades/months	Pr*WUP/100	Waste Produced
ARPTC	unitless	MAX(TIME, 0), 0)	Average GP
APr	Unidades	0), 0)	Average Production
ED	Unidades	SMTH1(Ords,AOT)	Expected Demand
St(t)	Unidades	St(t - dt) + (Pr - De) * dt	Stock
TP(t)	Unidades	TP(t - dt) + (PrR) * dt	Total Production
TW(t)	Unidades	TW(t - dt) + (WP - WU) * dt	Total Waste
WU	Unidades/months	TW*PWU/100	Waste Used
PrN	Unidades/months	W*Pt	Normal Production

Workers sub-model

RT	Months		0.2 Recover Time
DR	Per Month		0.5 Disabled Rate
PFCE	unitless		1.5 of Ethics
ATDEW	Months		3
HT	months		3 Hiring Time
T	Months		6 Expected Cost
PWD	unitless		6.75 Probability of Worker Disease
PWCA	unitless		10 Collective Agreements
UW	Unidades/(mo*Trabajador)		20 Units per Worker
WRT	months		54 Workers Rotation Time
PMW	unitless		59 Percentage of Male Workers
WS	US Dollars/(mo*Trabajador)		240 Women Salary
MFCEC	US Dollars/(mo*Trabajador)		250 Marginal FCE Cost
MWCAC	US Dollars/(mo*Trabajadores)		250 Collective Agreement
MS	US Dollars/(mo*Trabajador)		270 Men Salary
MODC	USD/(mo*Trabajador)		828 Marginal OD Cost
HN	Trabajadores/months	(DEW-TWr)/ HT	Hiring Need
TWr	Trabajadores	DW + W	Total Workers
DW(t)	Trabajadores	DW(t - dt) + (Inc - Recv) * dt	Disabled Workers
Recv	Trabajadores/months	DW/RT	Recovered
FCEC	US Dollars/months	FCE*MFCEC	FCE Cost
AFCE	unitless	Points(10): (0.000, 1.000), ...	Code of Ethics
ACAP	unitless	Points(10): (0.000, 1.0000), ...	Agreements to Productivity
APDP	unitless	Points(10): (0, 1.0000), ...	Diseases to Productivity
HIR	trabajadores/months	MAX(HN, 0)	Hiring
MSC	US Dollars/months	MW*MS	Men's Salary Cost
Inc	Trabajadores/months	OD*DR	Incapacitations
ODC	USD/months	OD*MODC	OD Cost

OD	Trabajadores	PWD/100*W	Occupational Diseases
DEW	Trabajadores	SMTH1(DP/Pt, ATDEW)	Desired Employee Workforce
TSC	US Dollars/months	SMTH1(MSC+WSC, T)	Total Salary Cost
WCA	Trabajadores	TWr*PWCA/100	Workers in Collective Agreements
Pt	Unidades/(mo*trabajadores)	UW*ACAP*APDP*AFCE	Productivity
W(t)	Trabajadores	Inc) * dt	Workers
WW	Trabajadores	W*(1-PMW/100)	Women Workers
FCE	Trabajadores	W*PFCE/100	Faults to the Code of Ethics
MW	Trabajadores	W*PMW/100	Men Workers
DIS	trabajadores/months	W/WRT+MAX(-HN, 0)	Dismissals
WCAC	US Dollars/months	WCA*MWCAC	WCA Cost
WSC	US Dollars/months	WW*WS	Women's Salary Cost

Emissions sub-model

TEF	Tons/Unidades	0.0001	Transport Emissions Factor
WEF	Tons/Unidades	0.0001	Waste Emissions Factor
PEFgr	Gr/Unidades	100	grams
grT	Gr/Tons	1000000	
TE	Tons/months	(DMD+IMD+De)*TEF	Transport Emissions
NAb	Tons/months	AbF*CO2_EitA	Natural Absorption
CO2_EitA(t)	Tons	NAb) * dt	CO2 Emissions in the Atmosphere
AbF	Per Month	CO2_EitA, 0)) Points(25):	Absorption Factor
PrEF	Tons/Unidades	PEFgr/grT	Production Emissions Factor
PrE	Tons/months	PrEF*Pr	Production Emissions
Emi	Tons/months	WE+PrE+TE	Emissions
WE	Tons/months	WEF*WP	Waste Emission